

26/WI.ECE-218-01 - Embedded Microcontrller Prjets

Project 2 - Automatic Headlight Control System

Team member:

Duy Hung, Dang

The aim of this project is to design a system to help student drivers learn about the ignition and headlight systems. The ignition system will indicate when the engine is ready to start, based on inputs from the seat and seatbelt sensors. If it is not configured properly, an alarm will sound. Once the engine has been started by pressing the ignition button, a red LED will turn on and the headlight system will activate. Drivers can configure the headlights to be on, off, or in auto mode, and they can also control the high-beam lights. Auto mode relies on a light sensor to detect the level of brightness

Starting repository: [hungd-union/Project1](https://github.com/hungd-union/Project1).

Summary of testing results

Ignition subsystem		
Behaviors	Test process	Results
When the driver sits down, display the message, “Welcome to enhanced alarm system model 218-W26”.	Power the board and monitor the board from VScode. Then, switch on the Driver Seat Sensor.	Driver welcome message activated.
Indicate ignition enabled with the green LED only when both seats are occupied and both seatbelts are fastened.	4 switch: DS, DSB, PS, PSB 1. No switch is on 2. Only Driver Seat Sensor is on 3. All switches are on	All tests passed 1. Alarm raised and all error messages printed. 2. Alarm raised and error messages for the driver's seatbelt, passenger seat and passenger seatbelt printed.

		3. The green LED is lit.
Start the engine when the ignition is ready and the ignition button is pressed.	Press the ignite button when the green LED is lit.	The green LED turns off and the red LED turns on. Display the message 'Starting the engine'.
Keep the engine running even if the driver or passenger should remove their seat belts or exit the vehicle.	Turn on the red LED. Then, turn off any DS, DSB, PS, PSB sensor (slide switch).	The red LED remains on.
When the engine is running, stop the engine when the ignition button is pressed.	Press the ignite button again when the red LED is on.	The red LED turns off.

Headlight subsystem		
Behaviors	Test process	Results
If the engine is running, and the user selects: ON, turn on both low beam lamps; if they select OFF, turn off all lamps.	Potentiometer connects to GND and 3.3V. 1. Wiper set to GND. 2. Wiper set to middle position. 3. Wiper set to VCC.	All tests passed 1. All lamps turn off. 2. All lamps turn on. 3. All lamps react based on photoresistor input. (Auto mode)
If the user selects the AUTO position: When the light level is above a “daylight” level, switch headlights off after a delay of 2 seconds. When the light level is below a “dusk” level, switch headlights on after a delay of 1 second. When between these two levels, the headlights should hold their previous state.	The LDR is connected in series with a 10 kΩ resistor, with the LDR connected to 3.3 V. Test it under a bright light and set the headlight to AUTO. 1. The sensor is covered by a hand at a close distance. 2. The sensor has a direct line of sight with the light source. 3. The paper is held approximately 7 inches above the sensor.	All tests passed 1. The light starts off. Then, after being covered for slightly more than two seconds, it turns on. 2. The light starts on. After around one second under bright light, it turns off. 3. The light remains as it was before.

Turn off all headlights when the engine is off.	Start the engine and set the lamps to the ON mode. Then stop the engine by pressing the ignition button again.	Both the low and high beams of the headlight turn off.
Automatic high-beam headlights (High-beam turns on along side with Low-beam headlights)	Set the headlights to the ON position. A two-position slider connects to 3.3 V and GND. 1. Turn the high beam on. 2. Turn the high beam off.	All tests passed. 1. The high-beam and low-beam lights are both on. 2. The high-beam lights are off while the low-beams are on.

P.S. I created my own version because my teammate was supposed to work on the ignition subsystem table, but he just copied the example from the assignment document.